Category A ledger: s-family, dimensions, Bridge C, T3 arithmetic, H3 orbit map

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Path shorthand: `SF/` = `/workspace/quantum-comp/reports/s_family_paths/`.

0. Standing hypotheses

- **(TB)** The manuscript theta lift u has type ρ_s and eigenvalue λ = 1/4. The λ = 1/4 constant-term criterion holds. Arb/python-flint ball arithmetic is correct. The rounding analysis of `SF/A2/scripts/certify_hybrid.py` is correct.
- **(F1–F5)** These are standard facts about Weil representations of finite quadratic modules, including Scheithauer's upper-triangular formula (as listed in `SF/A12_H1H2_odd`).
- **(I1–I5)** These are the local imports listed in `SF/A15_ST_balance` §1. They consist of F1–F5 together with the local models of `SF/A11_induction`.
- **Computer certificates.** "Exact mod-p rank" means the following. The defining equations have coefficients in Z[ζ][1/N], and they are reduced at a prime above p. Under reduction the rank can only drop. So a computed mod-p dimension is an upper bound for the dimension over C. Likewise, an integer bounded by ℓ/2 in absolute value is determined by its residue mod ℓ.

Notation: ρ_s is the s-family Weil representation, W is its γ_s- and (−1)-invariant subspace, E = ρ-span(δ₀) ⊆ W, and C = W ⊖ E is the cusp part. S, T and Z = S² are the standard generators. e(x) = e^{2πix}.

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1. s-family Weil representation ρ_s

1. Theorem. For every odd s ≥ 5, tr_C ρ(S) = tr_W ρ(S) = tr_E ρ(S) = 0. Hence dim C⁺ = dim C⁻ = d_cusp/2. Assume TB.

Proof. The matrix entries of ρ(S) lie in Q(ζ_n)·n^{−1/2}. For odd s, a Galois automorphism σ with σ(√n) = −√n fixes the label structure (family_s label matrices) and maps each of the spaces W, E, C to itself. It sends tr ρ(S) to −tr ρ(S). Since S² = Z acts as 1 on W, tr S is an integer combination of ±1, hence rational, so it is fixed by σ. A rational number equal to its own negative is 0. (`SF/A6_certify/REPORT.md`, `SF/galois_review/REVIEW.md` G1.)

2. Theorem. The same conclusion holds for s ≡ 4, 6 mod 8.

Proof. For these s, √n ∉ Q(ζ_{n/2}), so the automorphism σ of entry 1 still exists, and the argument applies verbatim. (`SF/galois_review` G1.)

3. Theorem. For every even s, tr_W S = 0.

Proof. There is a monomial unitary K_y on W that anticommutes with S and commutes with γ_s and −1. Then tr_W S = tr_W(K_y S K_y^{−1}) = −tr_W S. (`SF/A8c_Spairing_even/REPORT.md`. An exact check for even s = 6..60 is in `pairing_on_W.py`.)

4. Theorem. Assume F1–F5. Let s ≡ 0, 2 mod 8 and E = ρ-span(δ_0). Then tr_E S = 0 and tr_C S = 0, so dim C^+ = dim C^-.

Proof. By entry 8, tr_E S = ∏_p t_p, where t_p is the local trace on the local δ_0-span. Since s ≡ 0, 2 mod 8, m ≡ 0, 1 mod 4, so 4 | M, and the 2-adic exponent is a ≥ 2. There are two cases.

- **a odd, or u ≡ 1 mod 4.** Then $t_2 = 0$. For a ≥ 3 this follows from entry 5 (P-stability) and entry 6 (H2), by the lift lemma of entry 8, down to the base classes. For a = 2 the base classes (2, 1/4) ⊕ (8, −u/16) have trace 0 exactly when u ≡ 1 mod 4 (certificate: `SF/A11_induction/exact_E0.py`, exact arithmetic). This case does not use H2.
- **a even and u ≡ 3 mod 4.** The odd number u is a product of odd prime powers. Because u ≡ 3 mod 4, some p^e ‖ u satisfies p^e ≡ 3 mod 4. Then t_p = 0 by entry 7.

So tr_E S = 0. Since E ⊆ W (γ_s commutes with ρ and fixes δ_0), tr_C S = tr_W S − tr_E S = 0 by entry 3. (`SF/A12_H1H2_odd`, `SF/A31_entry7_check/REFEREE.md` §5.)

Corollary. Combining entries 1, 2 and 4: tr_C S = 0 for every s ≥ 5, with E = ρ-span(δ_0) when s ≡ 0, 2 mod 8.

5. Theorem (P-stability). Assume F1–F5. For every 2-adic exponent a ≥ 2, δ_h lies in the δ_0-span.

Proof. δ_h is obtained from δ_0 by the explicit word S T^{2^a} S. This is evaluated with Scheithauer's formula (F1–F5). (`SF/A12`.) An exact mod-p check (p = 33556993) for a = 2..7 and all 24 classes u mod 8 is in `SF/A13_H1_check/verify_h1.py`.

6. Theorem (H2). Assume F1–F5 and a ≥ 3. Then tr S = 0 on $(1 - P)E_0$.

Proof. The space X = $(1 - P)E_0$ has a basis indexed by $P^1(Z/M)$. The level is lcm(4, M), which is where a ≥ 3 is needed. In this basis S acts monomially and without fixed points, so every diagonal entry of S vanishes. (`SF/A12`.)

7. Theorem (odd-prime local factor). Let p be an odd prime, e ≥ 0, c a unit mod p^e, D_p = (Z/p^e, q(x) = cx^2/p^e), and ρ_p its Weil representation. Let E0_p be the $SL_2(Z)$-span of δ_0 in C[D_p]. Then:

- (i) E0_p is the space of even functions on D_p, of dimension (p^e + 1)/2.
- (ii) t_p := tr(S | E0_p) is equal to:
  - 1 if e is even;
  - (2/p) if e is odd and p ≡ 1 mod 4;
  - 0 if e is odd and p ≡ 3 mod 4.

  In particular t_p does not depend on c.

- (iii) For the global form D = ⊕_p D_p, E0 = ⊗_p E0_p, and tr_E S = ∏_p t_p.

Proof. Write γ(a) = Σ_{x mod p^e} e(ax²/p^e). Take S e_x = ε p^{−e/2} Σ_y e(−2cxy/p^e) e_y with ε = conj γ(c)/|γ(c)|. Then:

- tr S = conj γ(c) γ(−2c)/p^e and tr SZ = conj γ(c) γ(2c)/p^e.
- **e even:** γ(a) = p^{e/2} for every unit a, so tr S = tr SZ = 1.
- **e odd:** γ(a) = p^{(e−1)/2} (a/p) ε_p √p, so tr S = (−2/p) and tr SZ = (2/p).

By (i), the projection onto E0_p is ½(1 + Z), so t_p = ½(tr S + tr SZ) = (2/p)(1 + (−1/p))/2. This gives (ii). The opposite sign convention for T conjugates every quantity, and t_p is real, so it is unchanged. Part (i) is the odd case of H1/H2 (F1–F5). For (iii): $\delta_0 = \otimes\, \delta_{0,p}$; $SL_2(Z) \to \prod_p SL_2(Z/p^{k_p})$ is surjective; and ρ = ⊗ρ_p. Hence the span of the pure tensor δ_0 is the tensor product of the local spans. The phases are consistent because the total signature is 0 and every odd local part has even signature. (W and C are not tensor products, and the argument does not use such a decomposition.)

Certificate. `SF/A31_entry7_check/local_odd.py` covers all 53 odd prime powers p^e ≤ 201, each with c a residue and with c a non-residue. It computes t_p exactly in Z[ζ_{p^e}] modulo Φ_{p^e}, and dim E0_p as a rank mod $\ell \equiv 1 \bmod p^e$ (lower bound = dim W^+). It finds (i) and (ii) in all 106 cases.

8. Theorem (factorisation and lift lemma). Assume F1–F5. For even s, the original definitions of W, E and C reduce to C[D], with the complex-conjugate convention of `SF/A10`, so vanishing statements transfer between the two. Furthermore, tr_C S = −∏*p t_p, and tr*{P E0_a} S = t_{a−2} (2-adic lift lemma).

Proof. `SF/A11_induction/REPORT.md`. The factorisation is entry 7(iii) combined with entry 3. The lift lemma follows by comparing P E0_a with E0_{a−2} through the isotropic quotient. *Certificate:* the base classes a = 2, 3, 4, including P-stability, are computed exactly in `SF/A11_induction/exact_E0.py`.

9. Theorem. On the full space C[D], tr S = tr SZ = 0. The Eisenstein trace factorises as a product of local factors, and the local factor vanishes for every odd $p \equiv 3 \bmod 4$ with exponent 1.

Proof. tr S and tr SZ are normalised quadratic Gauss sums of the bilinear form, and they vanish by a one-line Gauss-sum evaluation. The factorisation follows from the tensor decomposition of entry 7(iii). The local vanishing is entry 7(ii) with e = 1. *Certificate:* 2-adic and odd local classes are checked exactly in `SF/A10_uniform_Spairing/exact_local.py`.

10. Certificate. Let $s \equiv 0, 2 \bmod 8$ with 8 ≤ s ≤ 74. Then tr_E S = 0, where E = ρ-span(δ_0).

Certificate: `SF/A31_entry7_check/global_trE.py`. It works on the full $D = A^{-1}Z^2/Z^2$. It computes the ⟨S, T⟩-span of δ_0 and tr_E S modulo two primes $\ell \equiv 1 \bmod 8M$. On E, $S^2 = 1$, so tr_E S is an integer of absolute value at most dim E < ℓ/2, and its residue determines it. All 18 values of s give 0. For s = 8, 16, 18, 24, 26, 32, 40, 48, 50, 56 and 58, an independent certificate of tr_C S = tr_E S = 0 is given by the invertible operator Y = Z − SZS, which anticommutes with S (exact mod-p rank; `SF/A8_Spairing_even/modp_trace.py`). For s = 10, C = 0.

11. Theorem (ST-balance, even s). Assume I1–I5. Let R = ST, so $R^3 = I$, and let $\omega = e^{2\pi i/3}$. For every even $s \geq 6$:

- $\mathrm{tr}_W R = -\omega$ if $s \equiv 2 \bmod 3$, and $\mathrm{tr}_W R = 0$ otherwise.
- $\mathrm{tr}_E R = -\omega$ exactly when (Exc) holds, and $\mathrm{tr}_E R = 0$ otherwise. Here (Exc) means: $3 \mid m - 1$, a is even, $u \equiv 1 \bmod 4$, and $v_p(M)$ is even for every prime $p \equiv 2 \bmod 3$.
- Hence $\mathrm{tr}_C R \in \{0, -\omega\}$. The R-eigenvalue multiplicities on C are (k, k, k) or (k+1, k, k+1).
- Moreover, $3 \mid d_{cusp} \Leftrightarrow \mathrm{tr}_C R = 0 \Leftrightarrow$ ($s \not\equiv 2 \bmod 3$ or (Exc)).

Proof. R splits into local factors because γ_s splits D (entry 13). The local traces r_p are computed in entry 13. The 2-adic base cases are certified exactly (`SF/A15_ST_balance/local_2adic_ab.py`). Since $R^3 = I$, the multiplicities $(n_1, n_\omega, n_{\bar\omega})$ satisfy $n_1 + n_\omega\,\omega + n_{\bar\omega}\,\bar\omega = \mathrm{tr}\,R$ with $n_1 + n_\omega + n_{\bar\omega} = \dim$. Solving gives the stated patterns. (`SF/A15_ST_balance/REPORT.md`.)

12. Certificate. For every even s with $6 \leq s \leq 400$ (198 values), $\mathrm{tr}_W R$ equals the Gauss-sum formula of entry 11. The method computes exact phase histograms, evaluates them at 40 digits, and rounds into the lattice $\mathbb{Z}[\omega]$. The lattice spacing is far above the error, so the rounding is exact. (`SF/A15_ST_balance/formula_even.py`, `gauss_even.py`.)

13. Theorem (local formulas for $\mathrm{tr}_W R$). Assume I5.

- γ_s splits D, since $(1/(m-1))\mathbb{Z}^2 \cap (1/p^k)\mathbb{Z}^2 = \mathbb{Z}^2$.
- For odd p, $r_p = \tfrac{1}{2}\,a_p(1 + \kappa_p)$. So $r_p \neq 0$ iff $\chi_{-3}(p^e) = 1$, and $r_3 \neq 0$ always.
- The case analysis mod 3 ($3 \nmid M$; $3 \mid m$; $3 \mid m - 1$) is exhaustive.
- Multiplicities of R computed mod p equal the true multiplicities for $p \nmid 3$, because $R^3 = I$ is diagonalisable mod p.

Proof. `SF/A15_ST_balance/REPORT.md` §§2–4. The splitting is the stated lattice intersection. r_p is a local Gauss-sum evaluation. The last point holds because $x^3 - 1$ is separable mod p.

14. Theorem (closed forms for d_{cusp}). Let G(m) denote the closed-form count of `SF/A`. Then:

- For odd m, $G(m) \leq (m+1)/2$, with equality iff $m = 1$ or m is a prime power.
- For odd $s \geq 5$, the closed form satisfies $d_{cusp} \geq (s-1)(s-3)/4 > 0$.
- $d_{cusp} = 0 \Leftrightarrow s \in \{6, 8, 10\}$.
- d_{cusp} is even for every $s \geq 5$.

These statements concern the closed-form expressions. They equal the true d_{cusp} wherever the closed forms are proved, namely $\dim V = s(s-1)$ for odd $s \leq 51$, and $d_2(k, v)$ for $k \leq 6$.

Proof. G is multiplicative; a prime power p^e contributes exactly $(p^e+1)/2$, and for coprime factors the product of the local values is strictly below $(m+1)/2$. The remaining statements follow by substituting into the closed forms (full proofs in `SF/A/REPORT.md`.) For odd s, the parity also follows from entry 1.

15. Theorem. Assume Chen's theorem (1973). Let $r = d_{cusp}/\dim V$ for odd s (formula level).

- Whether r is close to 1/4 depends only on the number k of prime-power factors of s(s−2). If every prime factor grows, then r → 1 − 3/2^k.
- liminf r = 1/4 is equivalent to the existence of infinitely many pairs (s−2, s) of odd prime powers.
- liminf r ≤ 5/8.

Proof. The first two statements follow from the multiplicative closed form of r: each factor tends to its limit as the prime grows. The third follows by applying Chen's theorem (p + 2 = prime or a product of two primes) to get k ≤ 3 infinitely often. (`SF/A3_chen/REPORT.md`.)

16. Theorem (isometry swap). Let h ∈ SO(A)(Q) with h(S^+) = S^- and Q∘h = Q. Then θ(R^+) = θ(R^-). At m = 7 (s = 98) such an h exists, so dim V(98) = 2406. Also, tr(g | Fix N) = 2m.

Proof. h maps the representation-number sets bijectively while preserving Q, so the two theta series agree term by term. (`SF/galois_review/REVIEW.md` G11.)

17. Theorem (Gauss-sum vanishing). |G(h/m)| is given by the factorisation of quadratic Gauss sums, and G(h/m) = 0 iff 2 ∥ m or 5 | m.

Proof. By the Chinese remainder theorem, G factors into local Gauss sums. Each local factor is evaluated in closed form; the factor at 2 vanishes exactly when 2 ∥ m, and the factor at 5 vanishes exactly when 5 | m (full computation in `SF/galois_review/REVIEW.md` G6.)

2. Dimensions of S(ρ_s)

Here S(ρ_s) denotes the space of Maass cusp forms of type ρ_s on W, of weight 0, with λ = 1/4. Every result in this section assumes TB. Upper bounds come from `SF/A2/scripts/certify_hybrid.py` (M = 6, Q = 40; Arb ball arithmetic). Lower bounds come from the sector lemma (entry 24) applied with d_cusp > 0, or from exact Hecke or T-row determinants.

18. Certificate. dim S(ρ_s) = 1 for s = 5, 7, 9, 12, 13, 14. (`SF/A/REPORT.md` §3, `SF/A2/REPORT.md`.)

19. Certificate. dim S(ρ_s) = 2 for s = 11 and s = 19. At s = 19 there is one form in each of two sectors. (`SF/A2`, `SF/A9_hecke_family/certify_hecke_s.py`.)

20. Certificate. Assume in addition Γ(1596)-invariance via Wohlfahrt. Then dim S(ρ_21) = 5. The lower bound comes from the Hecke images T′_p, p = 1597, certified in Arb at 256 bits from exact Fourier coefficients. (`SF/A7_s21/certify_hecke.py`.)

21. Certificate. dim S(ρ_23) = 3 (`SF/A6_certify`) and dim S(ρ_25) = 3 (`SF/A9`).

22. Certificate.

- 3 ≤ dim S(ρ_15) ≤ 4.
- 3 ≤ dim S(ρ_17) ≤ 4.
- dim S(ρ_s) ≥ 3 for s = 27, 29, 31.

(`SF/A9_hecke_family`.)

23. Theorem (Hecke commutation). If F is a cusp form of type ρ with eigenvalue λ, then T′_p F is again of type ρ with eigenvalue λ. T′_p commutes with the sector projectors, and its coefficients are given by the explicit formula of `SF/A7_s21`.

Proof. T′_p is a sum over double-coset representatives, which commute with the Laplacian. The ρ-twist is compatible with the representatives because ρ factors through the level (TB). (`SF/A7_s21`.)

24. Theorem (sector lemma). Let σ be a P-even sector other than all-plus, with P_σW ≠ 0. Then P_σu is a non-zero cusp form of type ρ_s with λ = 1/4.

Proof. P_σ commutes with ρ, so P_σu has type ρ_s and the same λ. Cuspidality and non-vanishing follow from the λ = 1/4 constant-term criterion (TB) applied in the sector σ; full proof in the source. (`SF/A6_certify`.)

3. Bridge C at s = 7

Here A_7 = [[2,7],[7,7]] and |D| = 35. ρ_{35} is the Weil representation of the discriminant form D(A_7) (S e_μ = (i/√35) Σ_ν e(−b(μ,ν)) e_ν, T e_μ = e(q(μ)) e_μ). χ is the generator of Hom(SL_2(Z), C^×) ≅ Z/12, with χ(T) = e(1/12) and χ(S) = −i.

25. Theorem. At s = 7, ρ_7|W = E ⊕ C with dim E = 36 and dim C = 6. Both E and C are irreducible, and dim C^+ = dim C^- = 3. Hence C has no 1-dimensional ρ-invariant subspace, and any S-eigenvector in C generates all of C as a ρ-module.

Proof. dim End(ρ_7|W) = 2 (exact mod-p upper bound; `SF/A32_35_to_rho7/exact_hom.py`), and E, C ≠ 0 are subrepresentations. So both summands are irreducible and non-isomorphic. The equality of C^+ and C^- is entry 1. (`SF/RTA_implications/REPORT.md`.)

26. Certificate. Assume TB. Then dim S(ρ_7) = 1. (`SF/A/scripts/certify_s.py`, `SF/A/REPORT.md` §3.)

27. Theorem.

- Q(√35) has class number 2, and 6 + √35 is a unit of norm +1.
- [[2,1],[1,−17]] is an odd lattice.
- There is no even binary lattice of determinant −35.

Proof. $6^2 - 35 = 1$, and the class number is standard. For an even binary form, the discriminant $b^2 - 4ac$ would equal 35, but 35 ≡ 3 mod 4 is not a square mod 4. (`SF/RTA_prereg/REPORT.md`.)

28. Theorem. Let C be any irreducible unitary representation.

- Every Γ-equivariant map C^{⊗a} ⊗ $\bar{C}$^{⊗b} → C is Z/3-graded by R-degree.
- If no T-exponents satisfy a + b + c − d ≡ 0 mod 1, then Hom_Γ(C⊗C⊗C, C) = 0.
- The products x⟨z, y⟩ and z⟨x, y⟩ are equivariant.

Proof. R^3 = I acts on the tensor product, and an intertwiner commutes with R. An intertwiner also commutes with T, so it can only connect T-eigenvectors whose exponents match. ⟨·,·⟩ is Γ-

invariant. (SF/A18_RTA_foundation/REPORT.md.)

29. Theorem. dim Hom_{SL₂(Z)}($\rho_{35} \otimes \chi^3$, $\rho_7|W$) = 1. A generator Φ is monomial: after rescaling, its non-zero entries lie in {±1, ±e(1/5)}. Its image is exactly C, and its kernel has dimension 29. Consequently C ≅ $\rho_5^- \otimes \rho_7^- \otimes \chi^3$, where $\rho_5^- \otimes \rho_7^-$ is the 6-dimensional subspace of ρ_{35} that is odd under both local reflections Z_5 and Z_7.

Proof and certificate.

- **Upper bound.** dim ≤ 1 from an exact mod-p rank computation at p = 29 401 and p = 14 729 401 (p ≡ 1 mod 29 400; SF/A32_35_to_rho7/exact_hom.py).
- **Lower bound.** The explicit 35 × 42 matrix Y with unit entries (SF/A32_35_to_rho7/X_chi3.npy) satisfies Y·S_{35}' = S_7·Y (210 of 210 entries) and Y·T_{35}' = T_7·Y (24 of 24 entries), exactly in Z[Z/29400] modulo Φ_{29400} (SF/A32_35_to_rho7/exact_verify.py). Generators suffice.
- **Rank and image.** The six image rows have disjoint column supports, so the rank is 6. The image is a 6-dimensional subrepresentation of W = E ⊕ C. By entry 25 it must be C.
- **Kernel and isomorphism.** The kernel contains the 11 labels with a zero local component and the 18-dimensional complement of the (−,−) sector on the 24 primitive labels, so Φ restricted to $\rho_5^- \otimes \rho_7^- \otimes \chi^3$ is an isomorphism onto C.

4. T3 and budget arithmetic

T3(n) is defined by:

- T3(n) = n/3 for n ≡ 0 mod 3;
- T3(n) = (4n + 2)/3 for n ≡ 1 mod 3;
- T3(n) = (2n + 1)/3 for n ≡ 2 mod 3 (in the integer form of SF/A20).

σ(n) is the number of steps needed to reach 1.

30. Theorem. With the integer form, T3 maps positive integers to positive integers. The expression (2n + 1)/3 is not an integer for n ≡ 2 mod 3; for example, (2·2 + 1)/3 = 5/3.

Proof. Check each residue class mod 3. (SF/A20, SF/A25.)

31. Theorem. $\sigma(3^k m) = k + \sigma(m)$, and $\sigma(3^k) = k$.

Proof. T3(3n) = n. Use induction on k. (SF/A23, SF/A24.)

32. Theorem. If σ(n) ≤ 539, then T3^{539}(n) = 1 when σ(n) is odd, and T3^{539}(n) = 2 when σ(n) is even.

Proof. T3 alternates 1 ↦ 2 ↦ 1. After reaching 1 at step σ(n), the remaining 539 − σ(n) steps alternate. (SF/A23_axiom01_revised/REPORT.md.)

33. Theorem. If σ(n) ≤ 540, then T3^{540}(n) = 1 iff σ(n) is even. In particular T3^{540}(4880) = 1, and among $n \le 10^6$, exactly 504 899 end at 1 and 495 101 end at 2.

Proof. As in entry 32. The counts are an exact enumeration (`SF/A30_540_thermo/tests540.py`).

34. Theorem. For every $j \geq 1$, the map $n \bmod 3^j \mapsto (T3^s(n) \bmod 3)_{0 \leq s < j}$ is a bijection from $\mathbb{Z}/3^j$ onto the words of length j over $\{0, 1, 2\}$.

Proof. Induction on j. On each residue class mod 3, T3 is affine with leading coefficient 1/3 or 4/3 or 2/3. So T3(n) mod 3^{j−1} is determined by n mod 3^j, and it runs over all classes mod 3^{j−1} as n runs over a fixed class mod 3. Exhaustive check for $j \leq 8$: `SF/A29_patterns_all/run_core.py`.

35. Theorem. T3 does not induce a map on $\mathbb{Z}/9$ (and more generally on $\mathbb{Z}/3^j$).

Proof. By entry 34, T3(n) mod 3^j depends on n mod 3^{j+1} and not on n mod 3^j.

36. Theorem. Each m has an explicit list of T3-preimages. Moreover, $T3(n) + 2 \leq (4/3)(n + 2)$, with equality iff $n \equiv 1 \bmod 3$.

Proof. The preimages are found by inverting each branch. For the inequality: $n \equiv 1$ gives equality, and the other branches give strictly less. (`SF/A24_threshold_1e18`.)

37. Theorem. $\lfloor (4880 - 21)/9 \rfloor = 539$.

Proof. 4859/9 = 539.89. (`SF/A20`, `SF/A23`.)

38. Theorem. $\lfloor 3^5 e^3 \rfloor = 4880$ and $\lfloor 35e^3 \rfloor = 702$.

Proof. $3^5 e^3 = 4880.84$ and $35e^3 = 702.99$, both evaluated with rigorous error bounds. (`SF/A19`, `SF/A25`.)

39. Theorem.

- $540 = 2^2 \cdot 3^3 \cdot 5 = 4860/9$.
- $\lfloor (4880 - f)/9 \rfloor = 540$ exactly for $f = 12, \ldots, 20$.
- $539.9 = 540(1 - 1/5400)$.
- 5399 is prime. Hence e^{2πi/539.9} = e^{2πi·10/5399} is a primitive 5399th root of unity, and its 539th power is not 1. Moreover, 539 steps of the phase e^{2πi/539.9} give 5390/5399 of a full turn.
- The beat period of 540 against 539.9 is 2 915 460.

Proof. Direct computation. Primality of 5399 is checked by trial division up to 73. (`SF/A25`, `SF/A30`.)

40. Certificate. Let m_{539} be the smallest n with $\sigma(n) = 539$. Then:

- $10^7 < m_{539} \leq 788\,638\,819\,521\,764\,360\,564\,859\,602\,047\,870$.
- For every $d \leq 291$ there is a seed below 10^{18} with $\sigma = d$.
- $\sigma(666\,954\,248\,808\,172\,552) = 291$.

The method is exact integer iteration (`SF/A24_threshold_1e18/exact_min.py`, `beam.py`).

41. Certificate (exact counts).

- max_{n ≤ 20 000} σ(n) = 53, attained at n = 16261.
- max_{n ≤ 50 000} σ(n) = 55, attained at n = 36043 and 36592.
- σ(4880) = 14 and σ(10^{18}) = σ(10^{18} + 1) = 92.
- σ(3^{539}) = 539.
- Every n ≤ 10^7 satisfies σ(n) ≤ 539.
- σ(3^{540}) = 540, and T3^{539}(3^{540}) = 3.

The method is exact integer iteration (`SF/A19_9maths_axioms/checks.py`, `SF/A23_axiom01_revised/budget.py`, `SF/A28`).

42. Theorem.

- $\varphi^7 = (29 + 13\sqrt{5})/2$.
- 1001 = 7·11·13.
- 18 + 1 + 520 = 539.
- 243 − 244 + 55 = 54.
- 4·8 + 5·12 = 92.
- 4·122 = 488.

Proof. Direct computation. For φ^7, use $\varphi^n = (L_n + F_n\sqrt{5})/2$ with $L_7 = 29$ and $F_7 = 13$.

43. Theorem. Consider any lift of T3 in which the integer coordinate evolves by T3 and steps are counted one-to-one. In any such lift, the number of steps to reach 1 is exactly σ(n).

Proof. The projection to the integer coordinate is a step-preserving semiconjugacy. (`SF/A20_T3_bulk_vs_shadow`.)

5. The H3 orbit map

Φ(n) = Σ_{s<539} w(r_s(n)) ∈ ⟨c⟩ ≅ Z/N, where r_s(n) = $T3^s$(n) mod 3, and w: {0, 1, 2} → Z/N.

44. Theorem. Φ is well defined for every n ≥ 1.

Proof. The sum is finite (539 terms) and every $T3^s$(n) is defined. (`SF/A28_H3_map_construction/REPORT.md`.)

45. Theorem (cocycle law). Let S_k(n) = Σ_{s<k} w(r_s(n)). Then S_{k+l}(n) = S_k(n) + S_l($T3^k$n). Hence w induces a monoid homomorphism {0, 1, 2}* → Z/N.

Proof. Split the sum at s = k.

46. Theorem (tail formula). If σ(n) ≤ 539 and L = 539 − σ(n), then Φ(n) = S_σ(n) + ⌈L/2⌉w(1) + ⌊L/2⌋w(2).

Proof. Apply entry 45 at k = σ(n). The tail orbit is 1, 2, 1, 2, …, with residues 1, 2, 1, 2, …. An exact check on 301 seeds × 9 weight choices is in `SF/A28_H3_map_construction/h3map.py`.

47. Theorem. Replacing the generator c by another generator multiplies Φ by a unit of Z/N. This preserves the zero set of Φ, its additive order, and τ. The manifolds M_A and M_B, with Z/N coefficients, give identical values.

Proof. Φ depends only on the cyclic group ⟨c⟩ ≅ Z/N.

48. Theorem.

- Tors $H^3(M_A; Z) = Z/N$, and there is no self-linking form on H^3 in dimension 7.
- For $M_B = L^7(N)$, $H^3(M_B; Z) = 0$ and $H^3(M_B; Z/N) = Z/N$.

Proof. Standard: universal coefficients, and the cellular chain complex of a lens space. A linking form in dimension 7 pairs H^3 with H^4, not H^3 with itself. (SF/A28.)

49. Theorem. Let $T = d + (\zeta(t) - 1)H_3\wedge$. Then $T^2 = 0$ iff $dt \wedge H_3 = 0$ and $dH_3 = 0$. A torsion class has no non-zero de Rham representative. On M_A, H_3 is exact, and T = e^{−fB} ∘ d ∘ e^{fB}; hence the T-cohomology is isomorphic to de Rham cohomology.

Proof. Expand T^2. de Rham cohomology is a real vector space, so torsion maps to 0. If $H_3 = dB$, conjugate d by the exponential. (SF/A25, SF/A28.)

6. Non-existence and vanishing results

50. Theorem. There exist positive integers n with σ(n) > 539. Specifically, σ(3^{540}) = 540.

Proof. Entry 31.

51. Theorem. For every k ≥ 38 with k ≠ 539, $\sigma(3^k) = k \neq 539$, and $3^k > 10^{18}$. Hence σ is not constantly equal to 539 on $n \geq 10^{18}$.

Proof. Entry 31, together with 3^{38} > 10^{18}.

52. Theorem.

- 3·61 + 162 + 223 = 568.
- 244 − 18 − 1 = 225.
- The alternating sum of the sequence of Betti numbers recorded in SF/A25 is −162.
- A closed oriented 7-manifold has $b_3 = b_4$. Hence no closed oriented 7-manifold has $b_3 = 223$ and $b_4 = 61$.
- With χ = 0 and rk = 3, χ/2 + rk = 3.
- $3^{18} < 10^9$.

Proof. Direct computation. $b_3 = b_4$ follows from Poincaré duality. (SF/A25.)

53. Theorem. 540·{2, 3, 4, 5, 10} = {1080, 1620, 2160, 2700, 5400}. None of these is an integer multiple of 539.9.

Proof. k·540/539.9 = 5400k/5399. Since 5399 is prime and does not divide 5400k for these k, the quotient is not an integer. (SF/A19, SF/A23.)

54. Theorem.

- $540/(1 + 1/540) = 291600/541 \approx 539.0018$.
- $540/(1 + 0.01 \cdot 0.185) \approx 539.0028$.
- $540(1 - 0.0001) = 539.946$.
- $540/(1 + \delta) = 539.9$ iff $\delta = 1/5399$.

Proof. Direct computation. (`SF/A30`.)

55. Theorem. Let $V \in \{\rho_{35}, \rho_{35}\}$, $U \in \{\rho_7|W, \rho_7|W\}$ and $k \in \mathbb{Z}/12$. Then $\mathrm{Hom}_{SL_2(\mathbb{Z})}(V \otimes \chi^k, U) = 0$ and $\mathrm{Hom}_{SL_2(\mathbb{Z})}(U, V \otimes \chi^k) = 0$, except in the two cases $(V, k, U) = (\rho_{35}, 3, \rho_7|W)$ and $(\rho_{35}, 9, \rho_7|W)$, where the dimension is 1 (entry 29 and its complex conjugate). In particular there is no non-zero intertwiner $\rho_{35} \to \rho_7|W$. All 12 T-eigenvalues of ρ_{35} occur in $\rho_7|W$, so the vanishing is forced by S, not by the T-spectra.

Certificate. The equations $XS_1 = S_2X$ and $XT_1 = T_2X$ are solved over $\mathbb{F}_p$ with $\zeta \mapsto$ a primitive 29 400th root of unity. This was done at $p = 29\,401$ and $p = 14\,729\,401$, with identical results. 96 Hom spaces were computed, and the mod-p nullity is 0 in each case listed.
(`SF/A32_35_to_rho7/exact_hom.py`, `run.py`, `results.json`.)

56. Theorem. For $s = 8, 10, 16$ and 18, no label-dilation Galois automorphism acts on $\rho_s(S)$ with sign -1.

Certificate. Exact enumeration of dilations (`SF/A8c_Spairing_even/galois_dilation_fast.py`).

57. Theorem. The class-number predictor $\mathrm{Cl}(K_s)$ of `SF/A5_classgroup` differs from $\dim S(\rho_s)$ at $s = 9$ and $s = 11$. The predictor $\mathrm{Cl}(4n_s)$ differs from it at $s = 23$ and 25.

Proof. The class numbers are exact (`SF/A5_classgroup/classgroups_A5.py`), and they are compared with the certified values of entries 18–22.

58. Theorem. At $s = 56$, the multiplicities of $R = ST$ on E are (187, 186, 187), so E is not R-balanced. On C they are (84, 84, 84).

Proof. Entry 11: at $s = 56$, (Exc) holds and $s \equiv 2 \bmod 3$. The dimensions are $\dim E = 560$ and $d_{cusp} = 252$, computed exactly (`SF/A15_ST_balance/exact_E0_R.py`).